UDACITY
Part of Accenture

Calculation of Basketball Scoring Probabilities

Data Scientist: Massimiliano Sermi

© 2024 Udacity.

Agenda

Results of basketball scoring calculations

- Overall Shooting Statistics
- Probabilities
- Conditional Probabilities - Future
- Conditional Probabilities - Retrospective

Overall Shooting Statistics

- Overall probability of a make 0.4899 (48.99%)
- Overall probability of a miss 0.5101 (51.01%)
- Overall proportion of three-pointers 0.5477 (54.77%)
- Overall proportion of two-pointers 0.4516 (45.16%)

Probabilities

- What is the probability of Steph making 3 of the next 4 shots?
  - 0.2399 (23.99%)
- What is the probability that 4 of the next 5 shots are three-pointers?
  - 0.2035 (20.35%)

- What assumptions can we make?
  - Fixed number of trials
  - Independent shots
  - Same probability each shot
  - Two outcomes (make/miss)

In reality, independence may be violated due to:

- Defense pressure
- Fatigue
- Game situation

Conditional Probabilities - Future

- Given the next shot is 3 pointer
  - Notation
    - P(3)=0.5477
    - P(Made ∩ 3) = 0.2293
    - P(Lead ∩ 3) = 0.2704
    - P(Lead | 3) = 0.4936
  - Probability he makes it
    - 0.2293 / 0.5477 = **0.4186**
  - Probability it was taken while his team had the lead
    - 0.2704 / 0.5477 = **0.4936**

- Given the next shot is 2 pointer
  - Notation
    - P(2)=0.4523
    - P(Made) = 0.4899
    - P(Made ∩ 2) = 0.2606
    - P(Lead ∩ 2) =0.2332
    - P(Lead | 2) = 0.5154
  - Probability he makes it
    - 0.2606 / 0.4523 = **0.5772**
  - Probability it was taken while his team had the lead
    - 0.2332 / 0.4523 = **0.5154**

Conditional Probabilities - Retrospective

Given Steph just made a shot

- ***Notation***
  a. P(Made)=0.4899
  b. P(Made∩3)=0.2293
  c. P(Made∩2)=0.2606

- Probability the made was a 3 pointer
  - 0.2293 / 0.4899 = **0.4680**
- Probability the made shot was a 2 pointer
  - 0.2606 / 0.4899 = **0.5320**
- Validity check
  - **0.4680 + 0.5320 = 1**

Data Interpretation:

When Curry make a shot , it is more likely to be a 2-pointer(53.20%) Than a 3-pointer (46.80%).